

Practical Setup :: Application Load Balancer (ALB) in EC2

□ prerequisites

- AWS account
- 2 or more EC2 instances (eg. Amazon Linux 2)
- Basic HTML server on each EC2 (eg. Apache or Nginx)

Step 1: Launch EC2 instances

1. Go to EC2 Dashboard > Launch instance
2. Name: webservr1 and webservr2
3. AMI: Amazon Linux 2
4. Instance Type: t2.micro (Free Tier)
5. Key Pair: Select or create new
6. Network settings:
 - select default vpc
 - select same Subnet or Availability zone if testing zone behavior.
 - Enable Auto-assign public IP
7. Security Group:
 - create or choose
 - Allow SSH (port 22)
 - Allow HTTP (port 80) — important!
8. Launch instances

Step 2 Install web server (Apache on both EC2s)

ssh into each ec2

```
ssh -i your-key.pem ec2-user@<Public-IP>
```

Then run

```
sudo su -
```

```
yum install httpd -y
```

```
service httpd start
```

```
cd /var/www/html
```

```
vi index.html
```

```
<h1> This is home page </h1>
```

On server 2 same steps.

Test via browser :-
http:// < IP for instance >

STEP 3 : create Target Group

1. Go to EC2 Dashboard → Load Balancers → Target Groups
2. Click create target group
 - Target type: instance
 - Protocol : HTTP
 - Port: 80
 - VPC: select default VPC
 - Name: myTarget-group
3. Health check path: / (default)
4. click next
5. Register Targets: select both EC2 instances and add to group
6. Click create target group.

STEP 4 : create APPLICATION Load Balancer (ALB)

1. Go to EC2 Dashboard → Load Balancers → create Load Balancer
2. select APPLICATION Load balancer
3. Name : myALB-1
4. Scheme : internet facing
5. IP Address type : IPV4
6. Listeners : HTTP (Port 80)
7. Availability Zones;
 - select vpc
 - choose at least 2 subnets (in different AZs ideally)
8. Click Next : Security Groups.

9. Security group

- create or select one with HTTP (port 80) access

10. Listeners and Routing:

- Listener : HTTP : 80
- forward to : myTarget-group

11. click create Load Balancer

wait a few minutes for provisioning

Step 5: Test the Load Balancer

1. Go to Load Balancer → Description
2. Copy the DNS Name of ALB (eg. myalb-123456, alb.amazonaws.com)
3. open in browser.

http://myalb-123456.elb.amazonaws.com

4. Refresh multiple times to see "webserver1" and "webserver2" alternating (Round Robin)

cleanup (Avoid charges):

- Terminate EC2 instances
- Delete Load Balancer
- Delete Target Group
- Delete Security Groups (if created)